

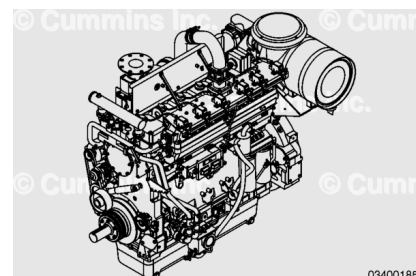
Trainings ▾

General Information

Note : The following procedure is **only** to be used for natural gas engines.

The valves **must** be correctly adjusted for the natural gas engine to operate efficiently. Valve adjustment **must** be performed using the values listed in this section.

Cummins Inc. has found that engines in most applications will **not** experience significant valve train wear after an initial adjustment is performed at the first 250 hours, then 750 hours. After this adjustment, it is recommended that the valves be adjusted every 1000 hours. Because valve train hardware is typically mixed between the cylinders during repairs, it is recommended that the valves be adjusted again 1000 hours after any service work that involves the valve train.



Standard Rating Valve Adjustment Limits

	mm	[inch]
Intake Valve	0.36	[0.014]
Exhaust Valve	0.81	[0.032]

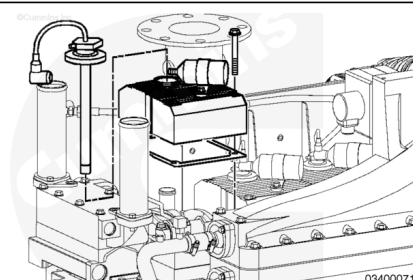
Catalyst Rating Valve Adjustment Limits

	mm	[inch]
Intake Valve	0.36	[0.014]
Exhaust Valve	0.97	[0.038]

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

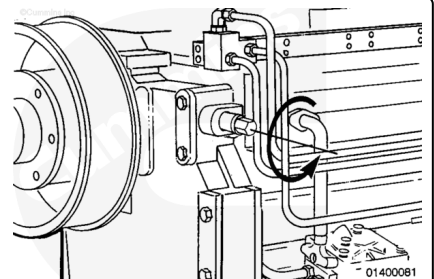


remove the negative (-) battery cable first and attach the negative (-) battery cable last.

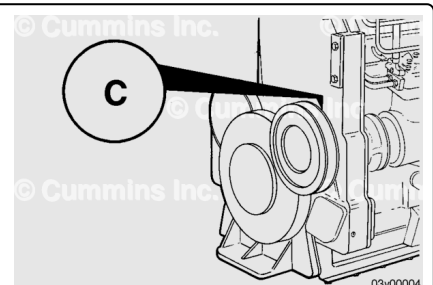
- Disconnect batteries. See equipment manufacturer service information.
- Remove the spark plug wire. Reference the KTA19 G CM558 Panel System Master Repair Manual, Bulletin 4021655. Refer to Procedure 013-014 in Section 13. (</qs3/pubsys2/xml/en/procedures/105/105-013-014.html>) Refer to Procedure 013-045 in Section 13. (</qs3/pubsys2/xml/en/procedures/105/105-013-045.html>)
- Remove the rocker lever cover (with ignition coil attached) and the spark plug adapter. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/18/18-003-011-tr.html>)
- Upon removal of the rocker lever cover, remove and discard the spark plug tube o-rings. Install new o-rings in the spark plug tubes and install the spark plug tube.

Adjust

Note : The barring device shaft turns approximately two revolutions before the engine begins to turn. The device will not turn the engine in the opposite direction of normal rotation.

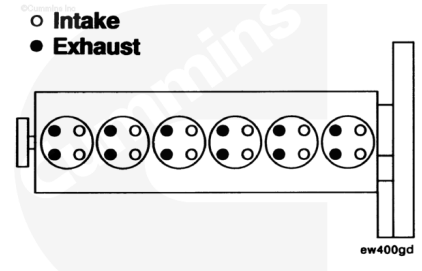


Push the shaft in and turn the barring device until the C mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



The illustration identifies the locations of the intake and exhaust valves.

Use the following procedure for the firing order of the engine.
Refer to Procedure 018-015 in Section V.
(/qs3/pubsys2/xml/en/procedures/18/18-018-015.html)



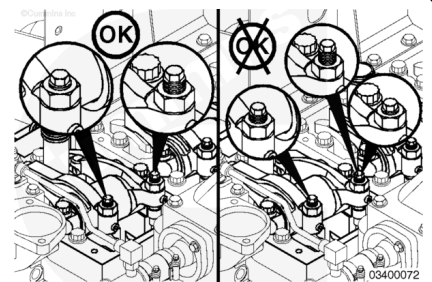
If the rocker levers have been removed, use this step to determine the cylinder to be set.

Note : All adjusting screws must be loose on all cylinders, and the push rods must remain in alignment.

Hold both rockers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.

Note : The push rods will be the same height above the top of the rocker lever housing on either cylinder number 1 or cylinder number 6. The cylinder that has its push rods at the same height is ready for valve adjustment.

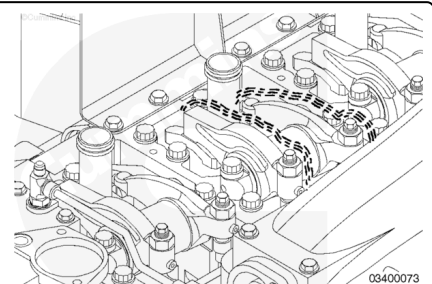
Note : The number of threads visible above the adjusting nut will not be the same. There will be more threads visible on the intake adjusting screw than on the exhaust.



Check the valve rocker levers on cylinder number 1 to see if both the intake and exhaust valves are closed. Both the intake and exhaust valves are closed when both levers are loose and can be moved from side to side. If both valves are **not** closed, rotate the engine 360 degrees and align the mark that is cast into the boss for the accessory drive seal on the front cover with the C mark on the accessory drive pulley.

When both rocker levers on cylinder number 1 are loose, set the lash on the following valves.

- Cylinder number 1 exhaust
- Cylinder number 1 intake
- Cylinder number 2 intake
- Cylinder number 3 exhaust
- Cylinder number 4 intake
- Cylinder number 5 exhaust.

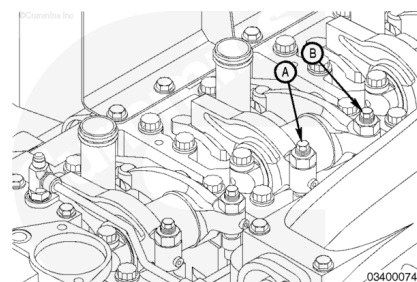


Standard Rating Valve Adjustment (Initial Set)

Reference Point	mm		[inch]
A	0.81	Exhaust	0.032
B	0.36	Intake	0.014

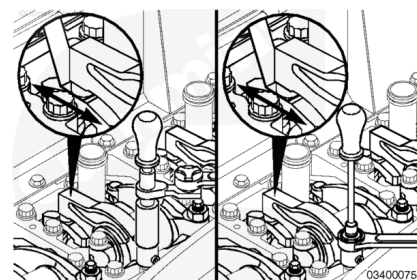
Catalyst Rating Valve Adjustment (Initial Set)

Reference Point	mm		[inch]
A	0.97	Exhaust	0.038
B	0.36	Intake	0.014



Use Cummins® valve setting gauge kit, Part Number 4919473, for the correct valve lash specification. Insert the gauge between the rocker socket and the crosshead.

Note : Make sure the feeler gauge is centered under the foot on the rocker lever.



Two different methods for establishing valve lash clearance are described below. Either method can be used. The torque wrench method has proven to be the most consistent.

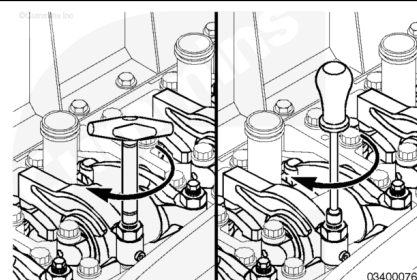
Torque Wrench Method:

- Use an Cummins® inch pound torque wrench, Part Number 3376592. Tighten the adjusting screw.

Torque Value: 0.68 n·m [6 in-lb]

Feel Method:

- Use a nut driver. Turn the adjusting screw **only** until the lever touches the feeler gauge.



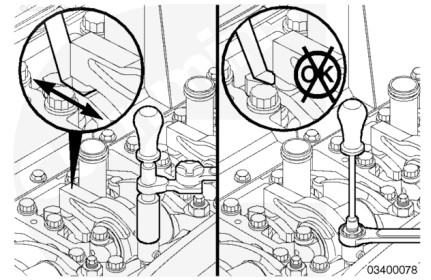
Valve Set - Torque Wrench Method:

- Make sure the parts are in alignment and squeeze the oil out of the valve train while tightening the adjusting screw.
- After loosening the adjusting screw at least one revolution, tighten the adjusting screw again.

Torque Value: .68 n·m [6 in-lb]

Valve Set - Feel Method:

- The feeler gauge **must** slide backward and forward with **only** a slight drag.
- Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker gauge will fit.
- Repeat the adjustment process until the clearance is correct on both the intake and exhaust valves.

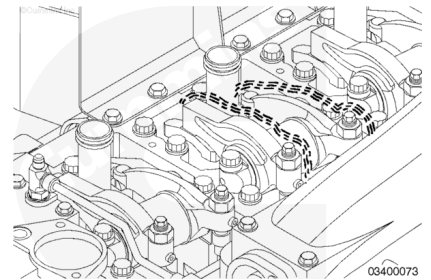


Note : The adjustment screw **must not** turn when the locknut is tightened.

Torque Value: 60 n•m [44 ft-lb]

Rotate the engine 360 degrees and align the mark that is cast into the boss for the accessory drive seal on the front gear cover with the C mark on the accessory drive pulley.

When both rocker levers on cylinder number 6 are loose, set the lash on the following valves.

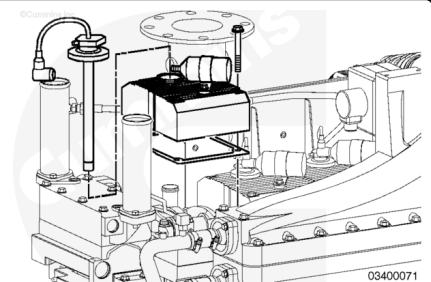


- Cylinder number 2 exhaust
- Cylinder number 3 intake
- Cylinder number 4 exhaust
- Cylinder number 5 intake
- Cylinder number 6 exhaust
- Cylinder number 6 intake.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the rocker lever cover (with ignition coil attached) and the spark plug adapter. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/18/18-003-011-tr.html>)
- Install the spark plug wire. Reference the KTA19 G CM558 Panel System Master Repair Manual, Bulletin 4021655. Refer to Procedure 013-014 in Section 13.

(/qs3/pubsys2/xml/en/procedures/105/105-013-014.html) Refer to Procedure 013-045 in Section 13.

(/qs3/pubsys2/xml/en/procedures/105/105-013-045.html)

- Connect batteries. See equipment manufacturer service information.
- Operate the engine and check for proper operation.